Madhav Anand Menon

+1 (646) 938-7079 | madhavanandmenon@gmail.com | linkedin.com/in/madhav-anand-menon | github.com/MadhavMenon10 | Personal Website

Education

University of Illinois Urbana-Champaign Expected May 2028

B.S. Computer Science and Physics, Minor in Mathematics (James Scholar Honors) **GPA: 3.97/4.0**

- Yee Seung Ng Award, Harold and Ruth Hayward Scholarship, Dean's List, Tau Beta Pi Honors Society
- Relevant Coursework: Honors Data Structures and Algorithms, Computer Architecture, Databases, Numerical Methods, High Frequency Trading Technology, Honors Linear Algebra, Probability Theory (Upcoming: Operating Systems)

Experience

Disruption Lab Feb. 2026 - Present

Software Engineer Champaign, IL, USA

- **Fall 2026: Working with the Google Cirq Team on partial Quantum Circuit rendering**
- Automated skill tagging across 100+ courses per API call, cutting manual effort 98%, by building a NACE-aligned skill extractor that returns deduplicated taxonomies.
- Supported 120+ concurrent users by shipping a FastAPI backend with custom token auth and dual-mode search (PostgreSQL full-text, pgvector cosine similarity).
- Enabled semantic search over Canvas course content by building a 5-stage RAG pipeline with 1024-dim embeddings via AWS Bedrock and sliding-window chunking.

Valeo Jun. 2026 - Aug. 2026

Software Engineer Intern Chennai, TN, India

- Cut IMU pitch error 37% against a survey-grade INS by engineering a 4-state Extended Kalman Filter that fuses gyro, accelerometer, and GPS velocity.
- Identified why the prior filter failed, with centripetal acceleration corrupting attitude 42 s before any trigger fired, by analyzing a 9,000-sample log.
- Cut false-rejection duty cycle 60% by tuning sensor-fusion gating against a 0.91 m/s^2 noise floor.

Algoverse AI Sep. 2025 - Jun. 2026

LLM Research Intern San Francisco, CA, USA (Remote)

- Dropped cross-layer detector ROC-AUC detectability from 1.000 to 0.634 by architecting DeployStega, an LLM-agent covert channel that embeds payloads in the authenticated permutation rank of real GitHub events.
- Cut detection from 83-99% to 8-23% TPR at 5% FPR across 200 preregistered runs against 4 capacity-matched baselines, using actor-, repository-, and text-disjoint splits to eliminate leakage.

DigiAlert Jun. 2025 - Aug. 2025

Software Engineer Intern Chennai, TN, India

- Achieved gap-free log ingestion under AWS's 2 req/s LookupEvents throttle by designing a boto3 CloudTrail pipeline with 5-min overlapping windows, EventId dedup, and retry backoff.
- Cut CSPM tooling cost 45% by building a Flask dashboard across 4 AWS services with CIS/NIST risk classification and dual-tier TTL caching.

Projects

GBDT Inference Engine | *CUDA C++, Python, XGBoost, NVIDIA RAPIDS Forest Inference Library*

- Cut median single-sample latency 3.9-6.8x below XGBoost's C API (24 µs vs 90-160 µs across 10 ensembles) by building a from-scratch C++/CUDA gradient-boosted decision trees inference engine.
- Beat NVIDIA RAPIDS FIL by 2.3-8.2x at batch size 1 , across 10 ensembles, by designing a dense-node tree layout and a thread-per-(sample, tree) kernel that fills full warps.
- Pinned the GPU/CPU crossover at batch size 1 and attributed 87% of large-batch cost to a 111:1 output asymmetry by profiling H2D/kernel/D2H over 30,000 seeded trials and building a two-pass harness.

CUDA-Accelerated Order Book Reconstructor | *CUDA, C++, CUB/Thrust, Nsight Systems*

- Processed 305M+ NASDAQ messages at 4.4M msgs/s on a cloud H100 by building an end-to-end CUDA order-book engine with a custom ITCH 5.0 decoder and warp-per-symbol reconstruction.
- Compressed 6.5 hours of backtesting into 83.8 s (279x), and 156x end-to-end over a single-threaded C++ baseline, by running 10,000 configurations in parallel.
- Reached 99.99997% reconstruction accuracy and eliminated a 27 GB tick buffer using a GPU open-addressing hash table and atomic-claim allocation over pinned SoA buffers.

Skills

Languages: C++, C, Python, JavaScript, TypeScript, SQL, Verilog, MIPS Assembly
Libraries and Frameworks: CUDA, PyTorch, FastAPI, Flask, pandas, scikit-learn, NumPy, Matplotlib
Tools: NVIDIA Nsight, Jira, Linux, AWS, LangGraph, LangChain, Git, Bash, TeX/LaTeX, Docker, Valgrind, MATLAB
Other Activities: Research at Parallel Programming Laboratory, CS 124 Project Manager, CodePath,TEDx Speaker